

# Basic SQL Operations in IBM Db2 DBMS Module 6

> **Definition:** SQL Operations in IBM Db2 adhere to ANSI/ISO SQL standards with specialized extensions for table creation, data manipulation, constraints enforcement, NULL handling, and complex JOIN queries.

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## 1. Detailed Technical Explanation & Executable Statements

### 1. Database Creation & Connection

```
```sql
```

-- Crea

```
-- Connect to the database
```

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### 2. Data Definition Language (DDL) & Constraints

```
```sql
```

-- Crea

```
-- Create STUDENT Table with Foreign Key & Check Constraints
```

CREA

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### 3. Data Manipulation Language (DML)

```
```sql
```

-- Inse

```
INSERT INTO STUDENT (STUDENT_ID, FIRST_NAME, LAST_NAME, EMAIL, AGE, DEPT_ID)
```

VALUE

```
-- Update Record
```

UPDA

```
-- Delete Record
```

DELE

```
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```

#### 4. NULL Value Handling in Db2

In Db2, `NULL` represents missing or unknown values. Standard comparison operators (`=`, `<>`) return `UNKNOWN` when evaluated against `NULL`.

```
```sql
```

-- Sele

```
-- Use VALUE() or COALESCE() to replace NULL with a default display value
```

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```
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```

#### 5. Join Operations in Db2

```
```sql
```

-- Inne

```
-- Left Outer Join
```

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```
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```

## 2. Core Concepts & Memory Keywords

- **PRIMARY KEY / FOREIGN KEY:** Relational integrity constraints.

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### 3. Must-Write Points for Exams

- Db2 supports standard `PRIMARY KEY`, `FOREIGN KEY`, `UNIQUE`, `NOT NULL`, and `CHECK` constraints.

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### 4. Quick Recall Flow

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